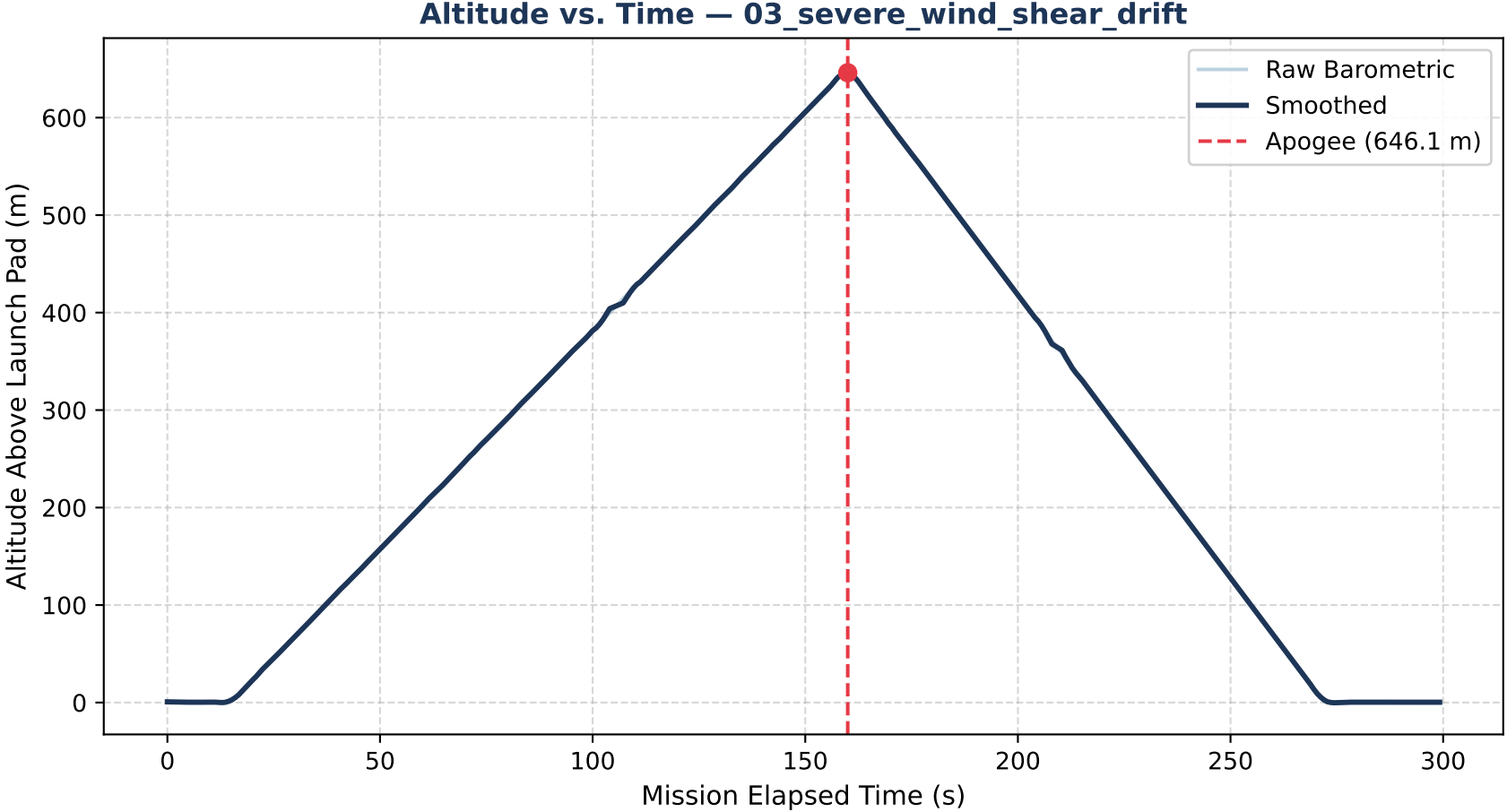
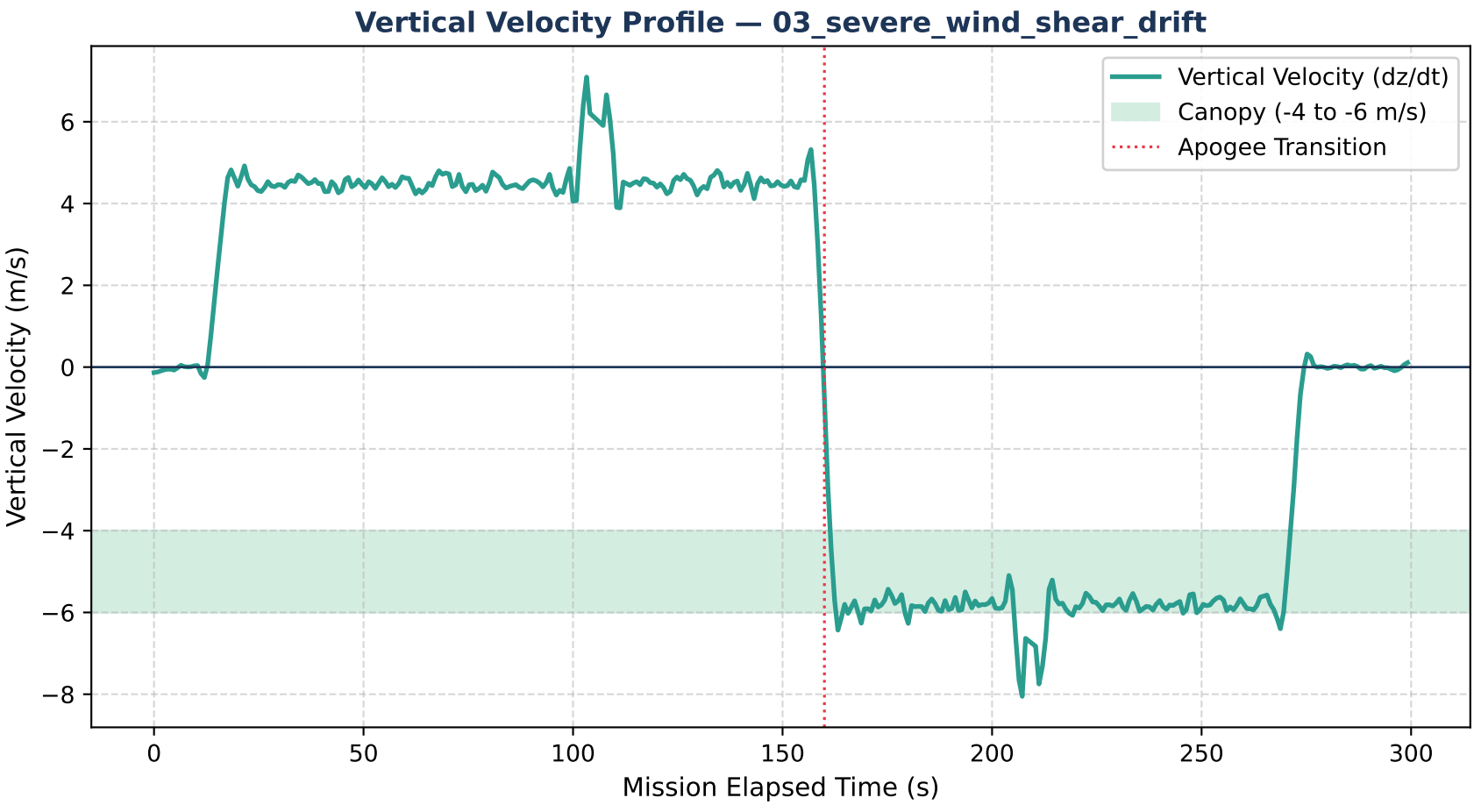
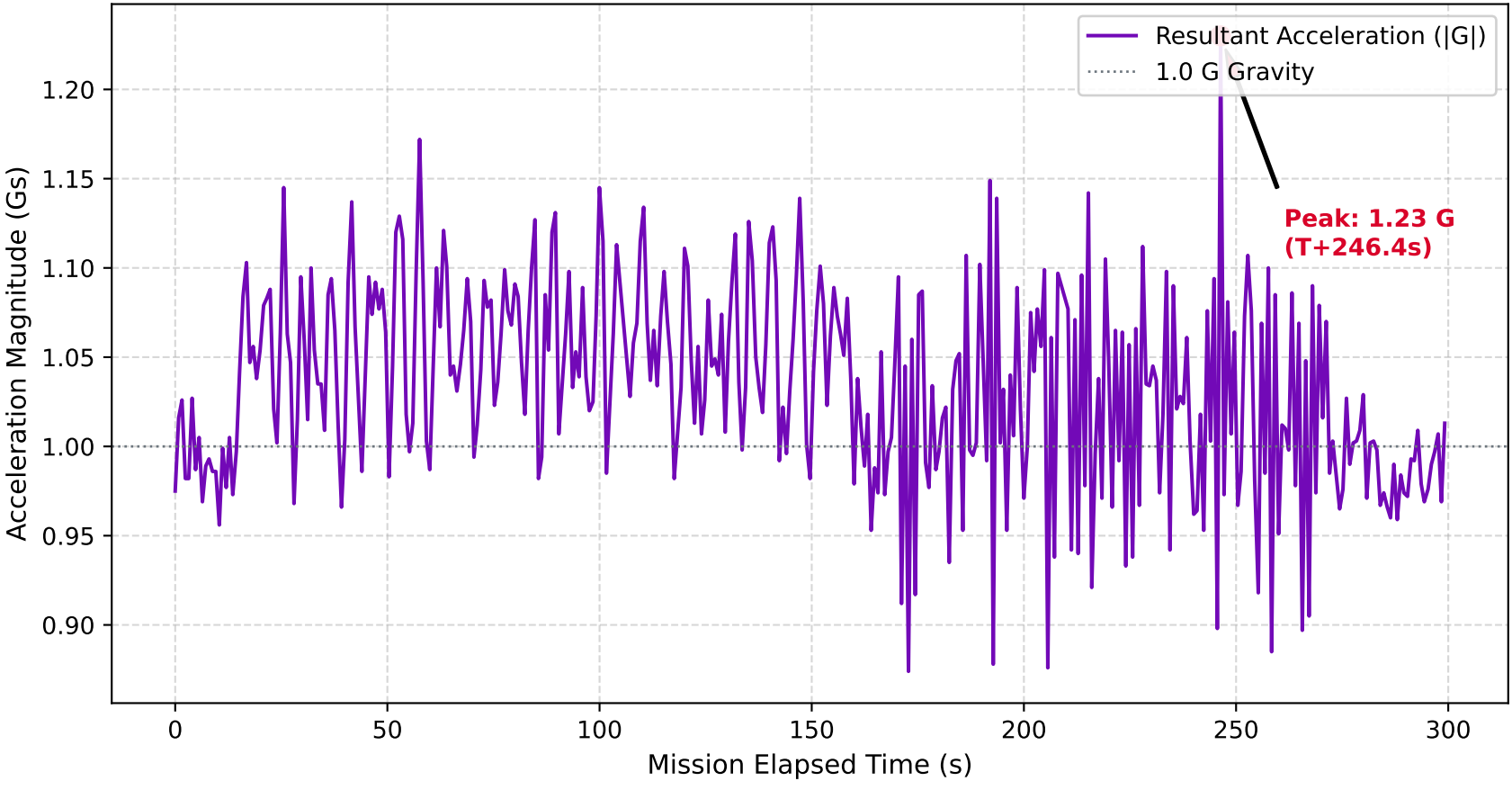
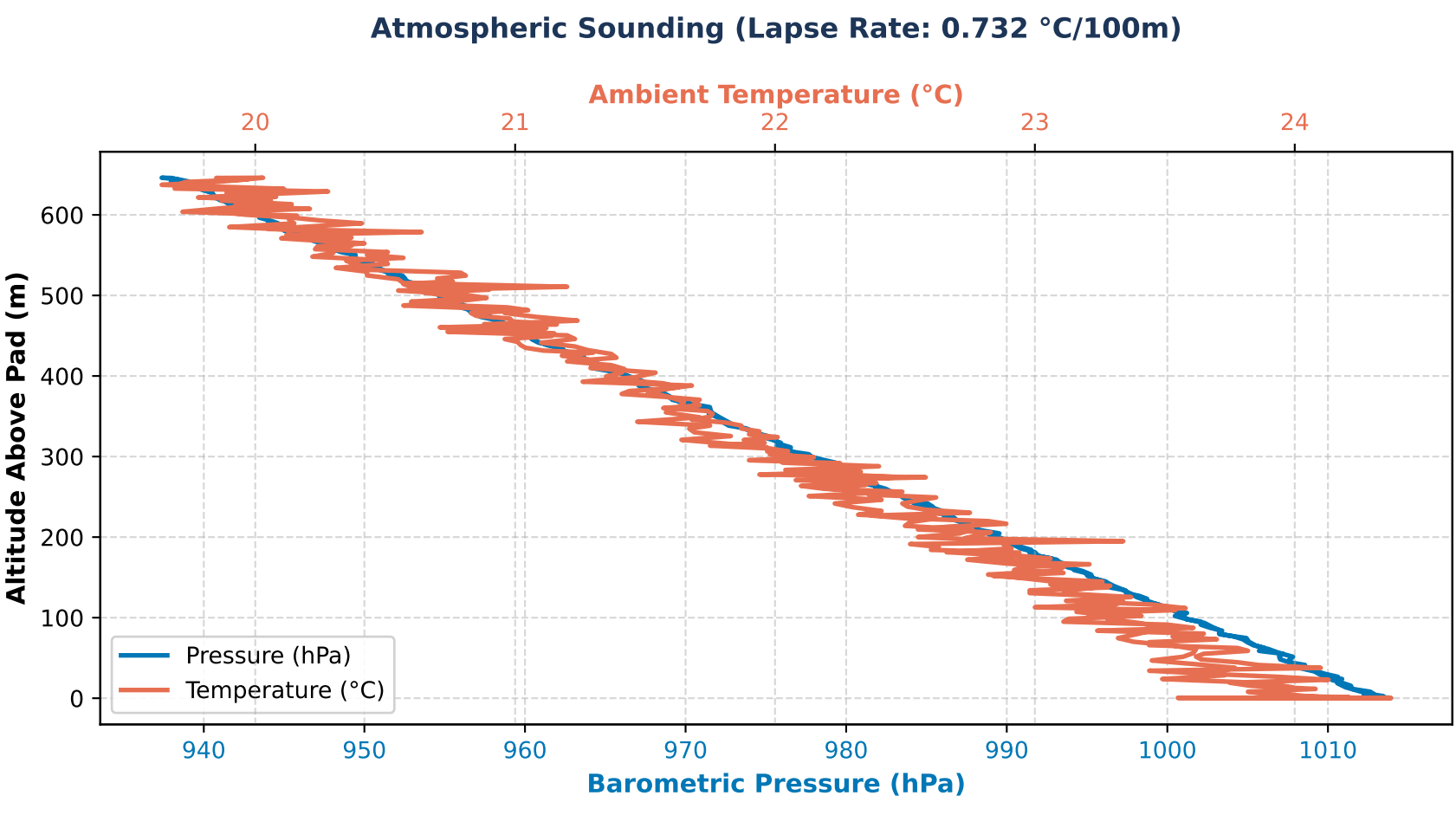
Cognitive CanSat - Mission Summary: 03_severe_wind_shear_drift

Flight Dynamics Parameter	Calibrated Value
Mission Name	03_severe_wind_shear_drift
Apogee (m)	646.06
Time to Apogee (s)	159.99
Max Ascent Velocity (m/s)	7.09
Max Descent Velocity (m/s)	-8.05
Max G-Shock (Gs)	1.23
Touchdown Shock (Gs)	1.1
Mission Duration (s)	299.17
Min Temperature (°C)	19.64
Max Temperature (°C)	24.37
Min Pressure (hPa)	937.38
Max Pressure (hPa)	1013.92
Lapse Rate (°C/100m)	0.732
Telemetry Frames	370
Landing Lat	22.581178
Landing Lon	88.384195


Altitude vs. Time — 03_severe_wind_shear_drift
Raw Barometric
Smoothed
Apogee (646.1 m)
Altitude Above Launch Pad (m)
0
100
200
300
400
500
600
Mission Elapsed Time (s)
0
50
100
150
200
250
300


Vertical Velocity Profile — 03_severe_wind_shear_drift
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
6
4
2
0
−2
−4
−6
−8
0
50
100
150
200
250
300
Mission Elapsed Time (s)


G-Shock & Acceleration — 03_severe_wind_shear_drift
Resultant Acceleration (|G|)
1.0 G Gravity
Peak: 1.23 G
(T+246.4s)
Acceleration Magnitude (Gs)
1.20
1.15
1.10
1.05
1.00
0.95
0.90
0
50
100
150
200
250
300
Mission Elapsed Time (s)


Atmospheric Sounding (Lapse Rate: 0.732 °C/100m)
Ambient Temperature (°C)
20
21
22
23
24
Altitude Above Pad (m)
600
500
400
300
200
100
0
Pressure (hPa)
Temperature (°C)
940
950
960
970
980
990
1000
1010
Barometric Pressure (hPa)

Cognitive CanSat Mission Dynamics: 03_severe_wind_shear_drift

